

Unary Systems: Thermodynamic conditions of equilibrium in closed and open systems. Gibbs energy-temperature-pressure diagrams and properties of substances. Scientific Group Thermodata Europe (SGTE) database. Condensed phase rule. Allotropy. Degree of transformation. (3 L)

Binary Systems with Two Phase Equilibria: Gibbs energy-composition diagrams. Solution models. Isomorphous systems. Calculation of liquidus and solidus in ideal systems. Lever rule. Iso-Gibbs energy curves. Equilibrium diagrams for systems exhibiting moderate departures from ideal behaviour. Congruent minimum and congruent maximum. Equilibrium and quasi-equilibrium solidification in isomorphous systems and corresponding microstructures. Stability of a phase. Stability function. Miscibility gap and chemical spinodal boundaries. Thermodynamic conditions and determination of miscibility gap, spinodal boundaries and consolute point. Decomposition inside and outside spinodal boundary and the microstructural features. Illustrative examples. (9 L)

Binary Systems exhibiting Invariant Reactions: Possible invariant reactions and the thermodynamic conditions. Eutectic systems. Primary and eutectic solidification and resulting microstructures. Ag-Cu, Al-Si and Pb-Sn systems. Eutectoid, monotectic and monotectoid systems. Peritectic systems. Solidification in peritectic systems and resulting microstructures. Cu-Zn system. Peritectoid, syntectic and metatectic systems. Ordered phases. Al-Cu and Ti-Al systems. Metastability. Iron-graphite and iron-cementite phase diagrams. Rules for construction of phase diagrams for complex systems. Typical errors and necessary corrections in the construction of phase diagrams. (12 L)

Ternary Systems: Graphical representation of ternary phase diagrams. Isothermal and vertical sections. Ternary isomorphous systems. Computation of liquidus and solidus boundaries. Congruent minimum and congruent maximum. Ternary eutectic systems. (6 L)

Computation of Phase Diagrams: CALPHAD methods and computation of multicomponent multi-phase equilibria and phase diagrams. (3 L)

Experimental Determination of Phase Diagrams: Isothermal methods. Microscopy, x-ray diffraction electrical resistivity and magnetic methods. Diffusion multiples. Non-isothermal methods. Thermal analysis and dilatometry. (6 L)

READINGS

4.1 TEXTBOOKS::

1. P. Gordon: Principles of Phase Diagrams in Materials Systems, McGraw-Hill, 1968.
2. S.P. Gupta, Phase Equilibria in Materials, Allied, 2003.
3. D.R.F. West and N. Saunders: Ternary Phase Diagrams in Materials Science, 3rd ed., Woodhead, 2002.
4. J.-C. Zhao (Ed.): Methods for Phase Diagram Determination, Elsevier, 2007.

4.2 *REFERENCE BOOKS::

1. D.A. Young: Phase Diagrams of Elements, Univ. of California, 1991.
2. H.A.J. Oonk and M.T. Calvet: Equilibrium between Phases of Matter, Springer, 2008.
3. M.H.G. Jacobs and H.A.J. Oonk: Equilibrium between Phases of Matter-Supplement, Springer, 2011.
4. F.N. Rhines: Phase Diagrams in Metallurgy, McGraw-Hill, 1956.
5. A. Prince: Alloy Phase Equilibria, Elsevier, 1966.
6. C.H.P. Lupis: Chemical Thermodynamics of Materials, North Holland, 1983.
7. M. Hillert: Phase Equilibria, Phase Diagrams and Phase Transformations, 2nd ed., Cambridge, 2007.
8. ASM Handbook Vol.3: Alloy Phase Diagrams, ASM International, 1992.
9. T.B. Massalsky (ed.): Binary Alloy Phase Diagrams, 2nd ed. (in 3 volumes), ASM International, 1990.